Abhinav Ganesh Vashishta

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.22
RV PU College	Class XII (Karnataka State)	2024	96.17%
Carmel	Class X (ICSE)	2022	93.20%

SKILLS

Programming Languages & Databases: C, C++, Python, SQL

Libraries & Frameworks: NumPy, Pandas, Pytorch, SciPy, Matplotlib, WebSocket

Engineering Capabilities: CAD (Fusion360, Blender), Fast Fourier Transform/Digital Signal Processing

Developer Tools: Git/GitHub, Linux (Command line), LaTeX, Jupyter Notebook, Agentic Workflow (Claude Code, Cursor)

PROJECTS

Real-Time Lead-Lag Arbitrage Engine *(Python, asyncio, WebSockets, NumPy/SciPy)*

- Built an FFT based Wiener-Khinchin cross-correlation engine (O(NlogN)) computing cross power spectral density to extract real time lead-lag and confidence between two markets after synchronizing time buckets
- Designed a lock free ring buffered ingestion layer with a Zero-Order Hold resampling to convert asynchronous multi-exchange ticks into a uniform discrete time grid
- Implemented a gated execution trigger and a latency-aware PnL backtester

Extended Physics-Informed Neural Networks (XPINNs) for Hyperbolic PDEs *(Python, PyTorch)*

- Solved a coupled hyperbolic conservation law system with discontinuous initial conditions by decomposing the domain into interface-coupled subnetworks
- Designed a composite loss function (PDE residual, Initial Conditions, interface-continuity and Boundary Condition terms) with weighted penalty terms, trained via batched Adam optimization over collocation, boundary and interface points
- Validated predicted spacetime evolution against the analytical solution, visualizing density/velocity profiles across the shock-like discontinuity.

Nonlinear Schrodinger Equation (NLSE) Solver for Fiber Optics *(Python, NumPy, Split Step Fourier Methods)*

- Implemented a second order accurate split-step Fourier solver from scratch (including custom radix-2/Bluestein FFT) to solve the NLSE, isolating dispersion and Kerr nonlinearity (SPM) via separate linear and nonlinear solvers
- Validated fundamental (N=1) and higher order (N=2) soliton dynamics against analytical predictions, demonstrating shape-preserving propagation and periodic breathing with energy conservation to a relative error under (10^-9)
- Characterized four propagation regimes (dispersion dominant, soliton, breathing, normal dispersion broadening) via automated test suite and cross solver comparison

Fast Automatic Frequency Calibration for PLL Frequency Synthesizers *(Verilog, Icarus Verilog, GTKWave)*

- Designed a binary search AFC scheme for PLL-based frequency synthesizers with digitally controlled VCOs, achieveing O(logN) convergence in under 8 iterations for an 8-bit control code
- Built a counter based frequency comparator immune to process/mismatch variation, with a two stage synchronizer for safe clock domain crossing between the reference and VCO clocks
- Verified via testbench simulation, locking a 200MHz VCO to a 100MHz reference with error under 1%, based on the AFC Scheme from Jeong et al. (IEEE RFIC 2009)

RELEVANT COURSES

Computer Science: Data Structures & Applications (with lab), Introduction to Programming (covering strong fundamentals), Artificial Intelligence (detailing training, implementation and use), Computational Physics

Physics & Mathematics: Advanced Mathematical Physics, Mathematical Physics, Introduction to Probability, Introduction to Statistics, Introduction to Group Theory, Differential Equations, Elementary Linear Algebra, Introduction to Group Theory, Maths for Physics, Calculus, Core Physics courses (Thermodynamics, Optics, Fluid Dynamics, etc)

POSITIONS OF RESPONSIBILITY

Overall Coordinator of the **Literary Society** of IIT Hyderabad

- Currently managing 5 clubs across the entire literary domain consisting of over 40 members and 5 heads

Coordinator of Event and Tech Management of Silhouettes, the **Fashion Club** of IIT Hyderabad

Head of the Writing and Poetry Club, under the **Literary Society** of IIT Hyderabad

- Led the team to podium finishes and victories in multiple inter-college competitions, conducted multiple competitions.

Coordinator of the LitFest under Elan & nVision, the **Techno-Cultural Fest** of IIT Hyderabad

- Organized multiple events with over 150 participating members including ideation, publicity and execution.

Coordinator at Rang De Manch, the **Dramatics and Speaking Arts Club** of IIT Hyderabad

Public Relations and Outreach Associate at FCC, the **Finance and Consulting Club** of IIT Hyderabad

- Supported the team in bringing multiple companies and influencers to IIT Hyderabad through effective communication

EXTRACURRICULAR

3D Modelling and Animation: Multiple projects completed with extensive knowledge and use of fluid simulations

Powerlifting & Fitness: Over 2 years of dedicated training with a passion for coaching others

Inter-IIT Cultural Meet: Represented IIT Hyderabad in over 3 events across the domains of performing arts and literature